3D print request

Option A · closed case

Team 6 · EXSU 620 · Project 06, CO_2 monitor

Submitted Monday 26 October 2026

All parts: PLA · 0.4 mm nozzle · 0.2 mm layers · **no supports needed** · do not scale (millimetres) · every file is already placed on the bed the right way up.
Files: each part comes as **.step** (use this) and as **.stl** (same shape, backup). The STEP solids are faceted (flat faces), so they slice exactly like the STL.
If time is short, please print in the order of the table: the T-piece first.

#	Part	File (.step + .stl)	Size (mm)	Colour	Settings	PLA
1	**T-piece (airway sampling adapter)** Breath flows through; the side spout feeds the sampling line	T6_tpiece_v1	32 × 28 × 60.0	Any	4 walls, 100 % infill (gas must not leak through), 5 mm brim	≈ 19 g
2	**Case base** Holds every module; the front panel carries the button, speaker grille and gas ports	T6_caseA_base_v1	152 × 94 × 50.0	White or light grey	3 walls, 20 % infill	≈ 105 g
3	**Case lid** Light window, ring holder, snap fit	T6_caseA_lid_v1	152 × 94 × 6.6	Same as base	3 walls, 20 % infill. Already upside down in the file	≈ 42 g
4	**Light cap** Diffuses the alarm light	T6_light_cap_v1	44 × 44 × 15.3	White or natural (translucent) PLA	2 walls, no infill needed. Open side up, as in the file	≈ 5 g
5	**Flow-cell clamp** Holds the CO2 flow-cell against the front panel	T6_cell_clamp_v1	49 × 41 × 8.5	Any	3 walls, 20 % infill	≈ 9 g
	Total				Roughly 13 h on a standard-speed printer; your slicer gives the real time	**≈ 180 g**

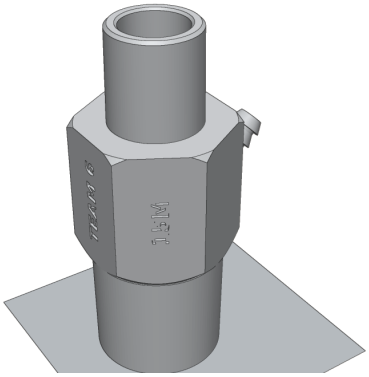
Why the T-piece needs 100 % infill: breath passes through it on the way to a CO_2 sensor, so it must be airtight. Its ends are standard 22 mm / 15 mm medical cone connectors (ISO 5356-1, 1:40 taper); please don't sand or scale them.
Light cap: printed in white or translucent PLA it glows evenly when the alarm light behind it turns on. Any other colour is fine for the other parts.
Questions: Abdelaziz, Team 6.



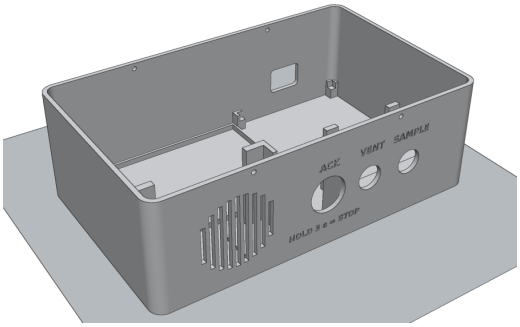
Parts as they sit on the bed

Option A · closed case · no supports on any part

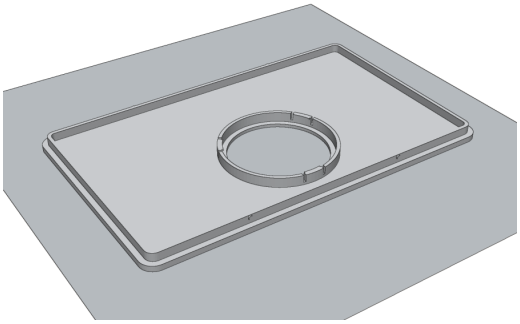
1 T-piece (airway sampling adapter) T6_tpiece_v1



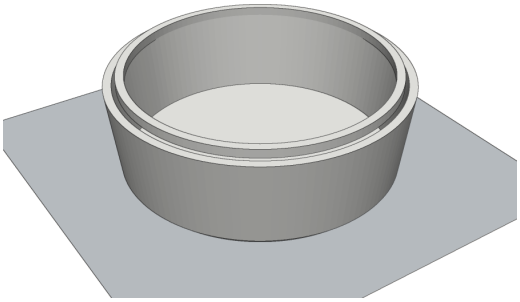
2 Case base T6_caseA_base_v1



3 Case lid T6_caseA_lid_v1



4 Light cap T6_light_cap_v1



5 Flow-cell clamp T6_cell_clamp_v1

